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Site-Selective sp^2 C−H Cyanation of Allenes via Copper-Catalyzed Radical Relay

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ABSTRACT: Compared with the extensively reported hydrogen atom transfer (HAT) at sp^3 C−H, abstraction of hydrogen atoms at the sp^2 carbon is extremely rare. Here, we communicate the site-selective cyanation of the sp^2 C−H bond of allenes using the strategy of copper-catalyzed radical relay. The reactions afford various allenyl nitriles directly from simple allenes with a broad substrate scope and a remarkable functional group compatibility under mild conditions. These reactions exhibit excellent site-selectivity toward sp^2 C−H, which can be attributed to the unique pocket created by the Cu-bound nitrogen-centered radical. The favorable HAT on sp^2 C−H is due to crucial hydrogen bonding between the fluoride bonded to the Cu(II) center and the hydrogen atom at the allylic position. These features enable the late-stage functionalization of druglike bioactive molecules containing an allene motif.

Allenes have been demonstrated as valuable building blocks extensively utilized in the synthesis of diverse intricate molecules.[1] The motifs are also found in natural products and pharmaceuticals, thereby showcasing their significance in biological systems.[2,3] For instance, because of their three-dimensional structure, allenic scaffolds have been recognized as the bioisosteres of propargylic compounds, furans, and spirocyclobutanes, which exist in the nucleic acids, a class of anti-HIV agents.[4] The allenic moiety in bioactive compounds (e.g., allenic steroids and prostaglandins) often acts as the reactive acceptor to inhibit the catalytic reactivity of enzymes (Figure 1A).[3]

Late-stage functionalization (LSF) of C−H bonds in bioactive molecules has emerged as a powerful tool for the advanced stages of their structural modification[5] to provide an efficient approach to adjust their biological and pharmacological properties. Therefore, this streamlined synthetic process of allene compounds would offer more efficient access to a wide array of structurally diverse allene compounds compared with the conventional synthetic approaches. However, very few examples of allenyl C−H functionalization were reported through the addition/elimination of C=C bonds[6] or Heck-type allenyl C−H alkenylation.[7] The rare example of sp^2 C−H alkenylation of allenes via palladium catalysis was reported by Carreira and co-workers with a directing-group strategy,[8a] and the elegant iron-catalzyed allenic C−H addition to imines and aldehydes was reported by Wang and co-workers.[8b,c] Therefore, the development of such general and practical methods is quite urgent. Herein, we report a copper-catalyzed sp^2 C−H cyanation of allenes where an uncommon site-selective hydrogen atom transfer (HAT) on the sp^2 C−H was observed as the key step, and the generated allenic sp^2 carbon radical was regioselectively quenched by a copper(II) cyanide.

In the last several decades, the catalytic C−H activation has garnered considerable attention from the scientific community because of its remarkable efficiency and versatility in molecular synthesis, and tremendous methods have been developed for the efficient transformation of various C−H bonds.[9] Among them, the methods involving hydrogen atom transfer (HAT) have been recognized as an effective approach,[10] and the reactions exclusively occurred to abstract sp^3 C−H bonds (Figure 1B-*i*), whereas the sp^2 C−H bonds are always retained (Figure 1B-*ii*).[11] Compared with the numerous studies on functionalization of sp^3 C−H bonds, however, the functionalization of the sp^2 C−H bond of allenes via HAT is not documented. Instead, because of the inherent property of cumulated diene, the reactions of allenes exclusively occur through the π-bonds breaking (e.g., radical addition).[12] Not to mention, multiple sp^2 allenic C−H bonds and active sp^3 allenylic C−H bonds often exist in the allene molecules simultaneously. Therefore, the site-selective abstraction of the hydrogen atom in sp^2 allenic or sp^3 allenylic positions becomes a challenge. Meanwhile, after the HAT occurs either in a sp^2 or sp^3 carbon center, the sequential regioselective functionalization of the sp^2 allenic C-radical and its resonant sp^3 propargylic C-radical (derived from sp^2 C−H), as well as the sp^3 allenylic C-radical and its resonant sp^2 vinylic C-radical (derived from sp^3 C−H), is another obstacle that results in the possible functionalization products on the allenic,[13a−e] propargylic[13f−h] or vinylic positions (Figure 1C). A synthetically useful protocol for the LSF of allene molecules would be needed to overcome these challenges and limitations.

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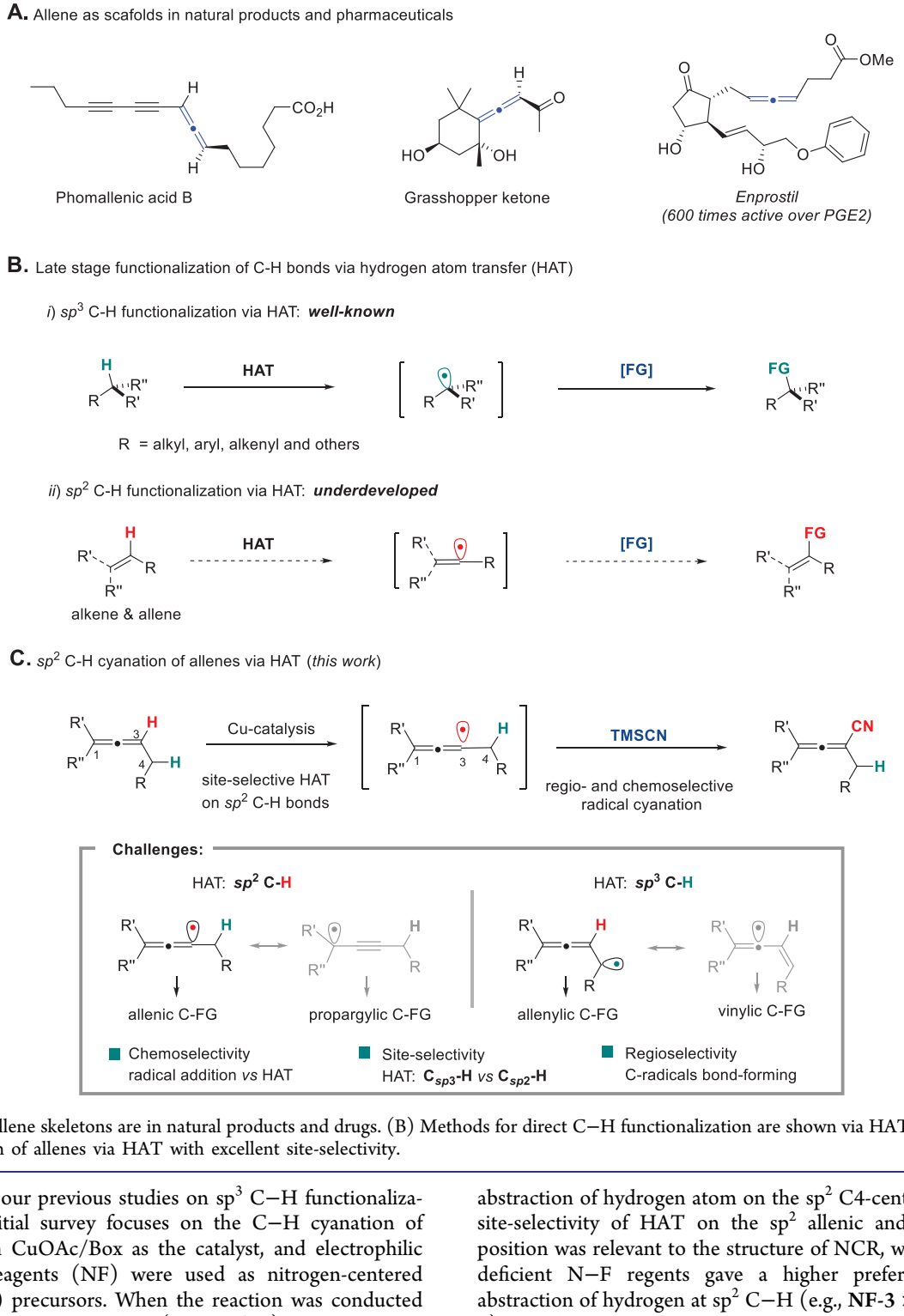
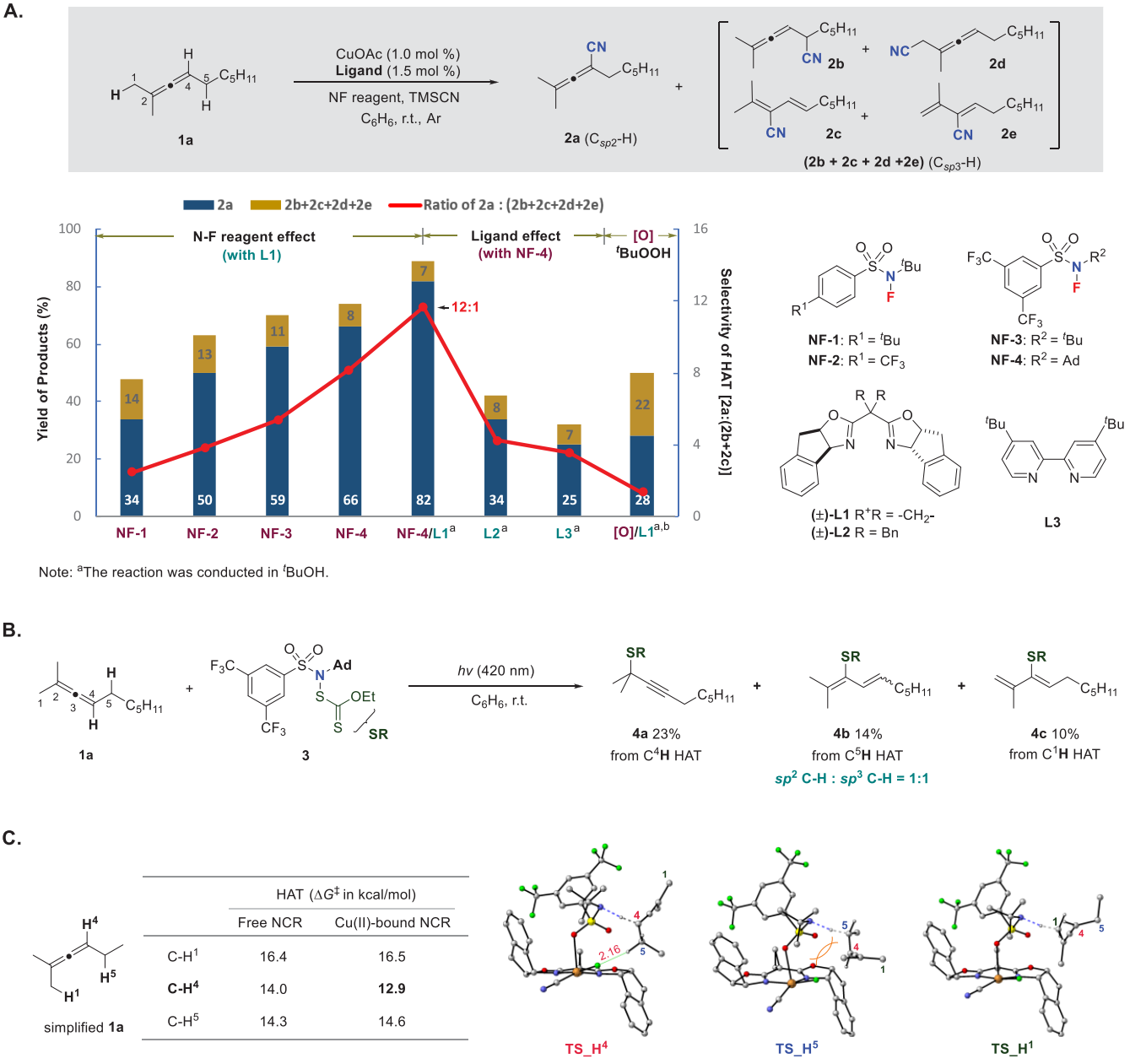


Figure 1. (A) Allene skeletons are in natural products and drugs. (B) Methods for direct C−H functionalization are shown via HAT. (C) sp^2 C−H functionalization of allenes via HAT with excellent site-selectivity.

Inspired by our previous studies on sp^3 C−H functionalization,[14] our initial survey focuses on the C−H cyanation of allene **1a** with CuOAc/Box as the catalyst, and electrophilic fluorinating reagents (NF) were used as nitrogen-centered radical (NCR) precursors. When the reaction was conducted with 1 mol % copper catalyst (CuOAc/**L1**) in benzene, as shown in Figure 2A, we were delighted to find that the direct C−H cyanation of allene **1a**, indeed, occurred via HAT rather than a radical addition pathway. More importantly, apart from our expected HAT on the allenylic sp^3 C−H at the C1 and C5 positions that delivered allenylic and vinylic cyanation products **2b**−**2e**, surprisingly, an unexpected HAT on the sp^2 C−H bonds occurred as a major reaction to deliver allenic nitrile product **2a**. We reasoned that the generated allenic carbon radical could be stabilized by the coplanar π-orbital of the adjacent C═C bond, which is attributed to the favorable abstraction of hydrogen atom on the sp^2 C4-center. Moreover, site-selectivity of HAT on the sp^2 allenic and sp^3 allenylic position was relevant to the structure of NCR, where electron-deficient N−F regents gave a higher preference for the abstraction of hydrogen at sp^2 C−H (e.g., **NF-3** > **NF-2** > **NF-1**), and the ratios of site-selectivity on allenic sp^2 C−H versus allenylic sp^3 C−H ranged from 2.4 to 5.4 (red line). By further changing the substituents on nitrogen from *tert*-butyl (**NF-3**) to adamantyl (**NF-4**), the ratio could be further improved to 8.3. Moreover, the reaction in *tert*-butanol as the solvent gave a better performance with a ratio of 12:1 so that the target product **2a** was obtained in 82% yield, along with the mixture of **2b**−**2e** in 7% yield. More importantly, further screening revealed that the site-selectivity was significantly influenced by the structure of ligands, whereas Box **L2** with a *gem*-dibenzyl group and bipyridine **L3** gave poor site-selectivity (sp^2/sp^3 =

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	HAT ($\Delta G^{\ddagger}$ in kcal/mol)	
	Free NCR	Cu(II)-bound NCR
C-H[1]	16.4	16.5
C-H[4]	14.0	**12.9**
C-H[5]	14.3	14.6

Figure 2. Site-selective C−H cyanation of allenes. (A) Reaction condition optimization with **1a**: screening of oxidants and ligands. Reaction conditions: CuOAc (1 mol %), ligand (1.5 mol %), TMSCN (0.8 mmol, 4.0 equiv), and NF (0.6 mmol, 3.0 equiv) in C_6H_6 (1.0 mL) under Ar at rt. The regioselectivity was detected by crude ^{1}H NMR. [a]Reaction in ^{t}BuOH (0.8 mL). [b]Reaction used ^{t}BuOOH as SET oxidant. (B) Metal-free reaction with free *N*-centered radical as the HAT species. (C) DFT calculation results.

4.3 and 3.5, respectively). Compared with the N−F reagents, when *tert*-butyl hydroperoxide (TBHP) was selected for the radical cyanation of C−H bonds, where the free *tert*-butoxy radical was recognized as a key species to carry out the HAT process,[15] an even worse site-selectivity of HAT (sp^2/sp^3 = 1.3:1) was observed, thereby indicating the unique reactivity of **NF-4** on the site-selective sp^2 C−H cyanation. A similar site-selectivity of HAT (sp^2/sp^3 = 1:1) was also obtained under conditions of light-induced chain reaction where a free NCR was generated from N−S reagent **3** (Figure 2B).

DFT calculations were conducted by using simple **NF-3** along with **L1**, where the excellent selectivity (C_{sp2}/C_{sp3} = 9:1) was still observed in ^{t}BuOH. As shown in Figure 2C, the DFT results proved that the Cu(II)-bound NCR exhibits a higher selectivity on the sp^2 over sp^3 C−H bonds than the free NCR. Compared with abstracting hydrogen atom at C1 (**TS_H^1**) and C5 (**TS_H^5**), an obvious hydrogen bonding interaction (2.16 Å) was observed between the fluorine in the Cu(II) center and the hydrogen atom at the allylic position of the substrate, which results in the lower energy barrier of **TS_H4**, thereby facilitating the highly site-selective HAT on the sp^2 C4 center.

With established reaction conditions, the substrate scope was then investigated. As shown in Figure 3a, various *tri*-alkyl-substituted allenes were demonstrated as the suitable candidates for the radical cyanation of sp^2 C−H bonds. Although the statistic number of hydrogen atoms on the sp^3 allenylic position is much higher (typically 6−8 times) than that of sp^2 allenic positions, the reaction always afforded allenic nitriles **5**−**8** with excellent site-selectivity (9:1 to 17:1) and

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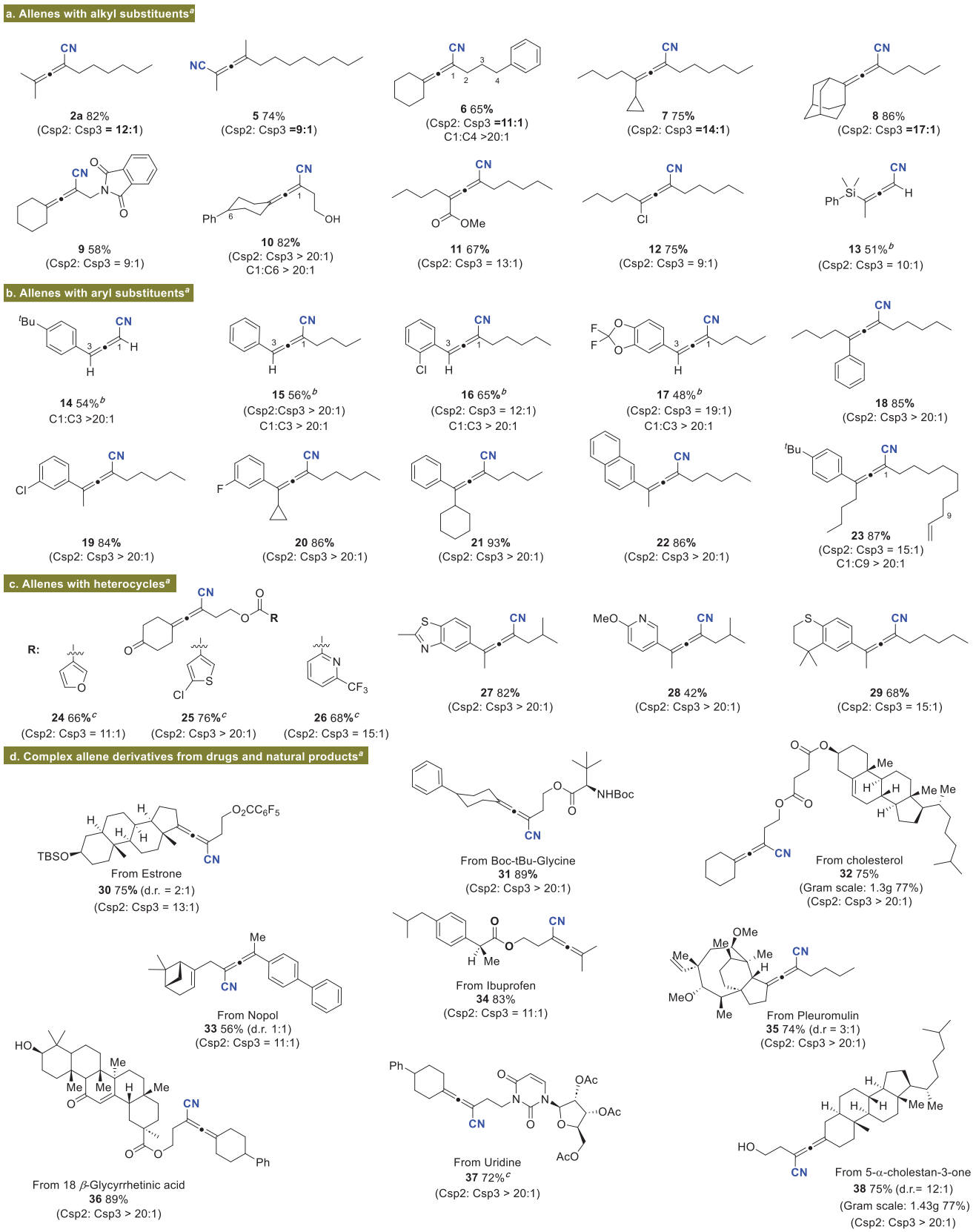


Figure 3. Substrate scope for sp^2 C−H cyanation. [a]The reaction was conducted at 0.2 mmol scale: allene substrates (1 equiv), CuOAc (1 mol %), (±)-**L1** (1.5 mol %), TMSCN (2.0 equiv), **NF-4** (1.5 equiv) in ^{t}BuOH (0.8 mL) under Ar at rt for 15 h. The isolated yields were given, and the site-selectivity (C_{sp^2}:C_{sp^3}) was determined by crude ^{1}H NMR. [b]**NF-5** (2.0 equiv) and TMSCN (3.0 equiv). [c]With the standard conditions, a second portion of TMSCN (0.2 mmol) was added, **NF-4** (0.1 mmol), and the reaction was stirred for an additional 15 h. **NF-5**: *N*-fluoro-*N*-adamantylbenzenesulfonamide.

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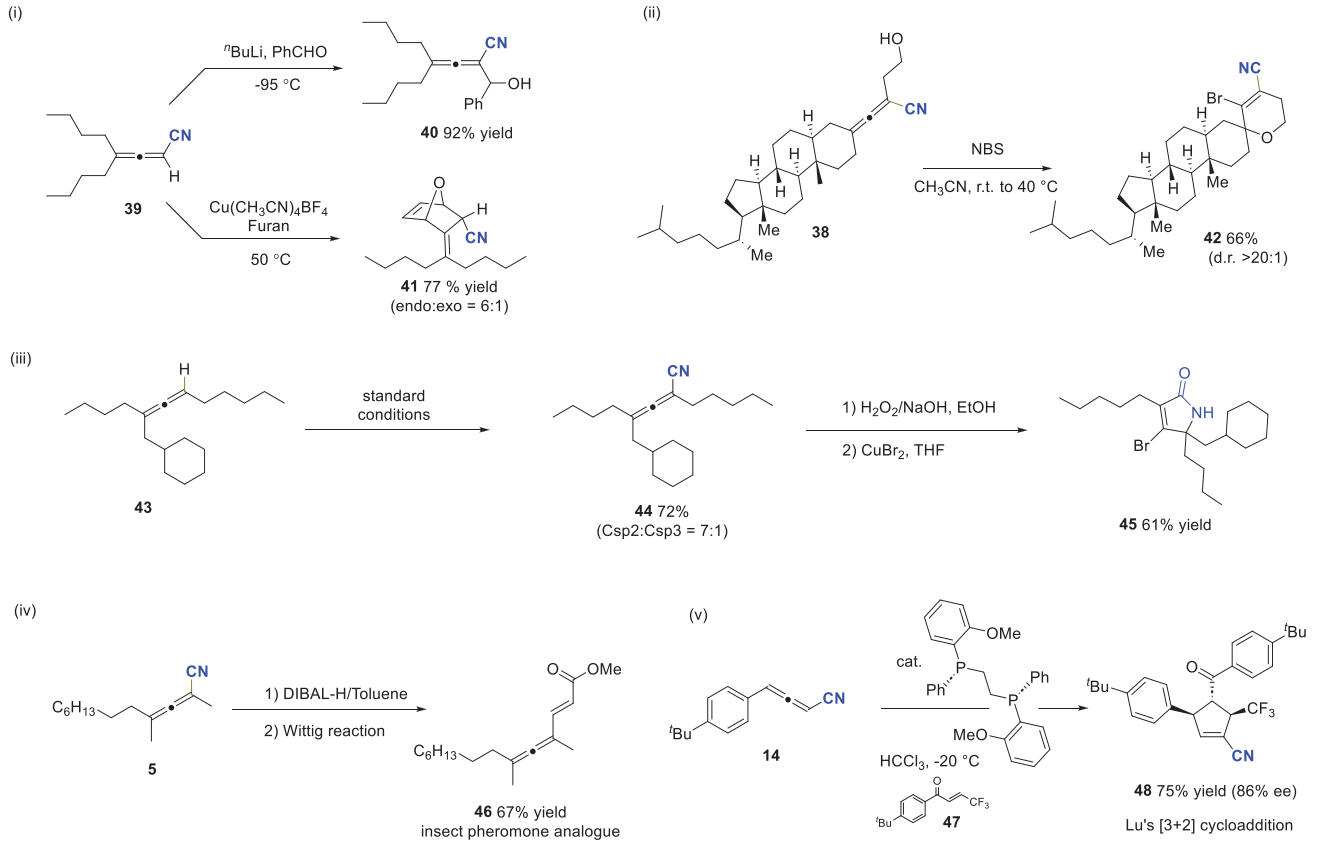


Figure 4. Synthetic applications and transformations.

good yields (65−86%). More importantly, for the allene substrate bearing extra benzylic C−H bonds, the reaction occurred predominantly at the allenic position to provide desired product **6** where the benzylic C−H was intact. Allenes bearing synthestically versatile functionalities, such as imide, hydroxy, carboxylate, and halide (e.g., Cl), in one allenic position are suitable for the reaction to deliver products **9**−**12** with decent results. In addition, 1,1-disubstituted allene was also suitable for the sp^2 C−H cyanation to deliver product **13** in 51% yield with good site-selectivity.

As shown in Figure 3b, we were delighted to find that aryl-substituted allenes provided even better performance where an excellent site-selectivity (typically $C_{sp^2}/C_{sp^3} \approx 20:1$) was consistently observed in products **15**−**23**. More importantly, for the allene substrates bearing two types of allenic sp^2 C−H bonds, the reactions also exhibited excellent site-selectivity. For instance, the reaction of monosubstituted aryl−allene proceeded smoothly to deliver **14** in 54% yield where the HAT process only occurred to the allenic C1 position with excellent site-selectivity. Moreover, 1,3-disubstituted aryl−allenes were also identified as efficient substrates for the sp^2 C−H radical cyanation where the reactions also exclusively occurred at C1 to give **15**−**17** in moderate yields with an excellent site-selectivity (C1/C3 > 20:1). It should be emphasized that for the allene substrates bearing an additional sp^3 allylic C−H bond (**23**), the highly site-selective radical cyanation of sp^2 C−H bond was also observed, where the allylic C−H was intact. Finally, various heterocycles, such as furan (**24**), thiophene (**25**), pyridine (**26** and **28**), benzothiazole (**27**), and cyclic thioether (**29**) (Figure 3c) are also compatible, which did not erode the site-selectivity of the reactions.

Subsequently, the investigation was focused on complex allene substrates derived from drug molecules and natural products. As shown in Figure 3d, we were delighted to find that a series of complex substrates derived from estrone (**30**), glycine (**31**), cholesterol (**32**), nopol (**33**), ibuprofen (**34**), pleuromulin (**35**), glycyrrhetinic acid (**36**), uridine (**37**) and cholestanone (**38**) were suitable for this sp^2 radical cyanation with excellent site-selectivity (typically $sp^2/sp^3 > 20:1$), where an array of functional groups, including ester, ether, amino acids and sugar moieties, are compatible to the standard conditions. Again, other active C−H bonds in benzylic (**31**, **36**), allylic (**32**), and adjacent carbon positions of heteroatom (**30**, **35**−**36**, **38**) were retained. In addition, the practical application of our method in the laboratory was demonstrated by the gram-scale preparations of compounds **32** and **38** with excellent performance.

Allenyl nitriles have been recognized as important building blocks for organic synthesis. For instance, compared with the unactivated allenes, the lithiation of allenic nitrile **39** and the subsequent 1,2-addition with benzaldehyde readily occurred to give allenylic alcohol **40** in 92% yield; the D−A cyclization reaction of **39** with furan also readily gave **41** under the copper catalysis (Figure 4i). In addition, for the unprotected allenic alcohol **38**, the electrophilic brominating cyclization proceeded very well to give **42** in 66% yield, in which the vinyl bromide moiety allows for further cross coupling reactions (Figure 4ii). Moreover, with the standard sp^2 C−H radical cyanation, the generated product **44** could undergo the sequential oxidative hydrolysis and copper-mediated cyclizative bromination to provide **45** in 61% yield where the allene moiety could survive under the oxidative conditions (Figure 4iii). Again, tetrasubstituted allenic nitrile **5** could be used for the synthesis of the

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analogue of insect pheromone **46** via a sequential reduction of the nitrile group with diisobutylaluminium hydride (DIBAL) and Wittig reaction (Figure 4iv). Finally, the nature of electron-deficient allenyl nitrile allowed the interesting asymmetric Lu's [3 + 2] cyclization,[16] which yielded chiral cyclopentenyl nitrile **48** in 75% yield and 86% ee where chiral phosphine was employed as the organocatalyst (Figure 4iv).[17,18]

Finally, the enantioselective cyanation of sp^2 C−H was also investigated (Figure 5). With our previous catalytic system

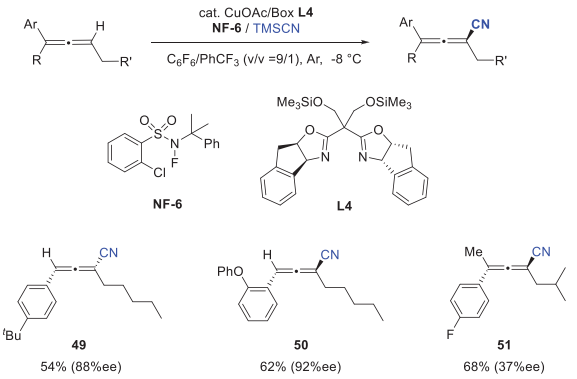


Figure 5. Enantioselective cyanation of sp^2 C−H bonds.

Cu(I)/Box **L4**,[13b] the reactions of 1,3-disubstituted allenes **49** and **50** gave enantioselectivities of 88% and 92% ee, respectively, although the asymmetric control on the sp^2 C−H cyanation of trisubstituted allenes (**51**, 37% ee) needs further attention.

In conclusion, a highly site-selective sp^2 C−H radical cyanation of allenes has been established via a copper-catalyzed radical relay. This method offers a streamlined approach for the synthesis of multisubstituted allenic nitriles from simple allenes with high efficiency. This catalytic system is also suitable for the late-stage functionalization of allene-based bioactive compounds

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.3c11368.

Experimental procedures and characterization data (PDF)
DFT calculation data (XYZ)

Accession Codes

CCDC 2300479 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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